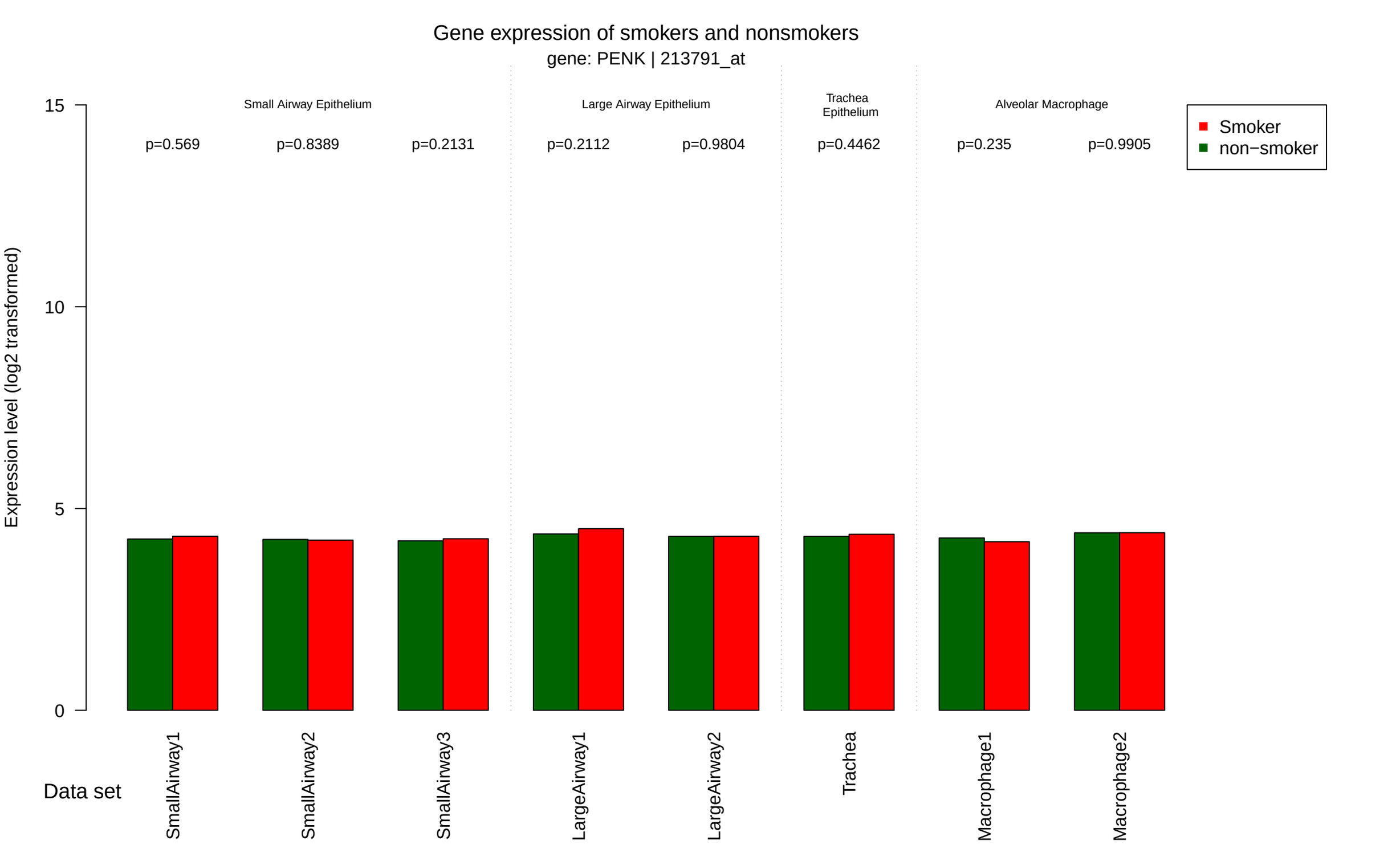
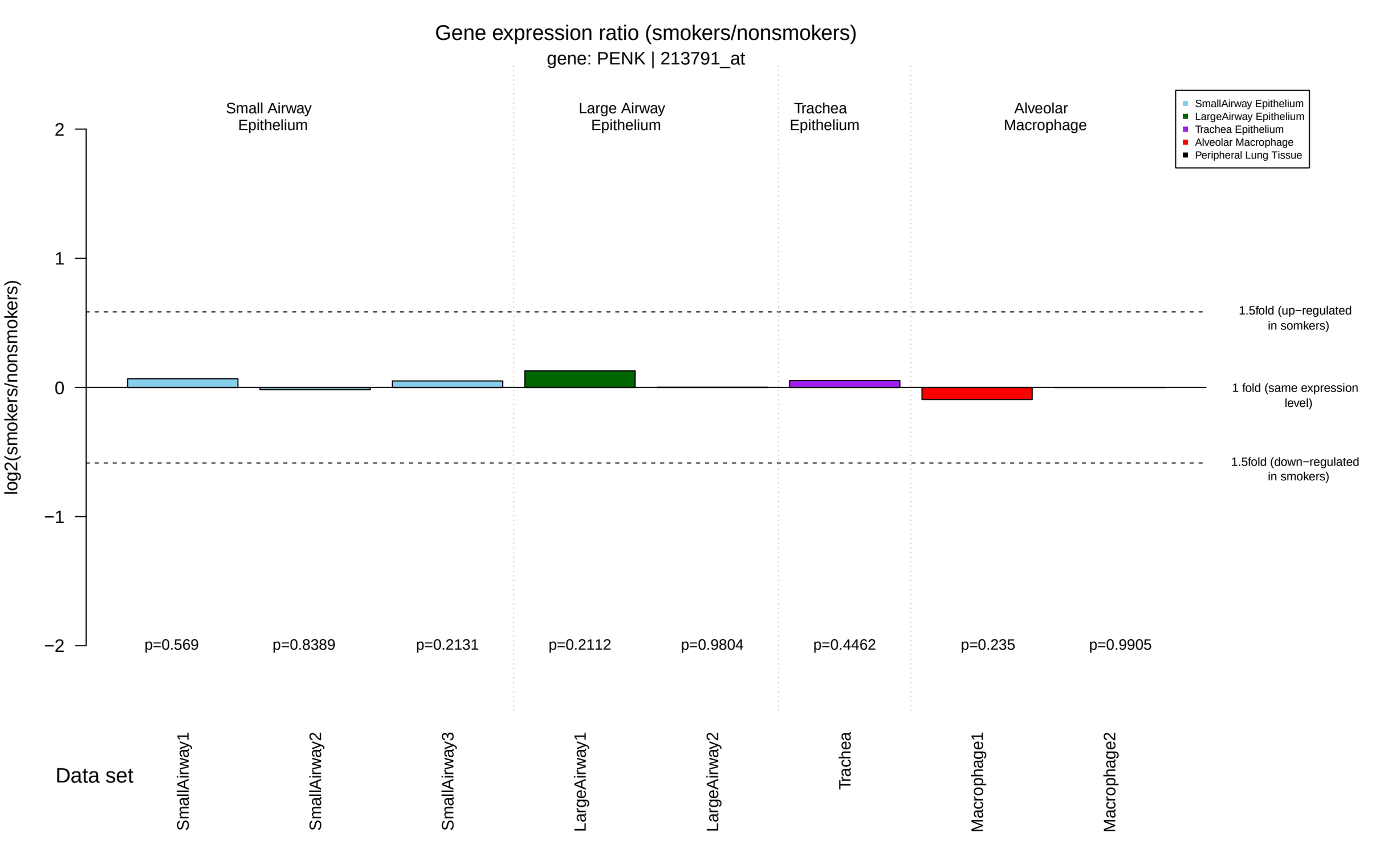
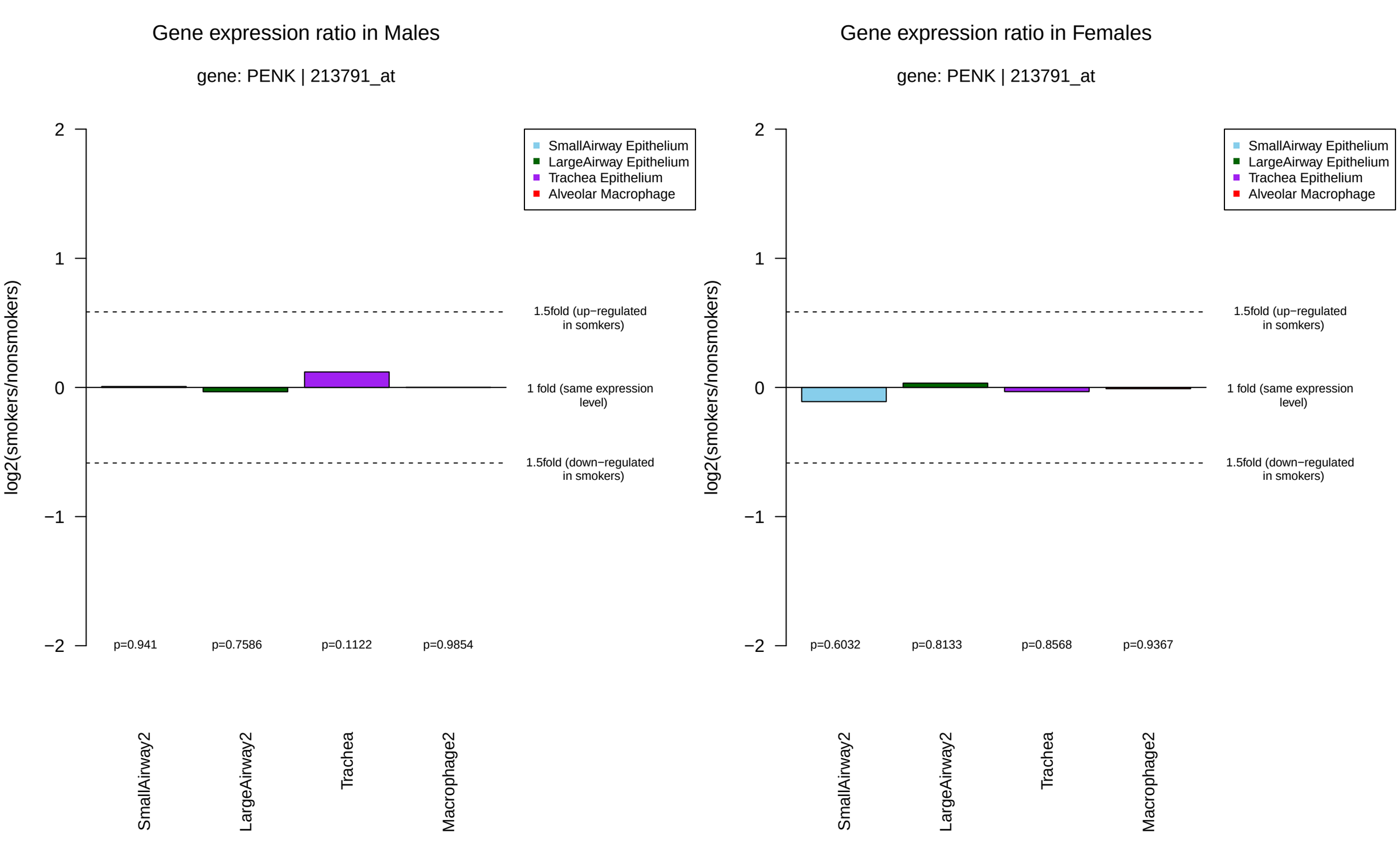
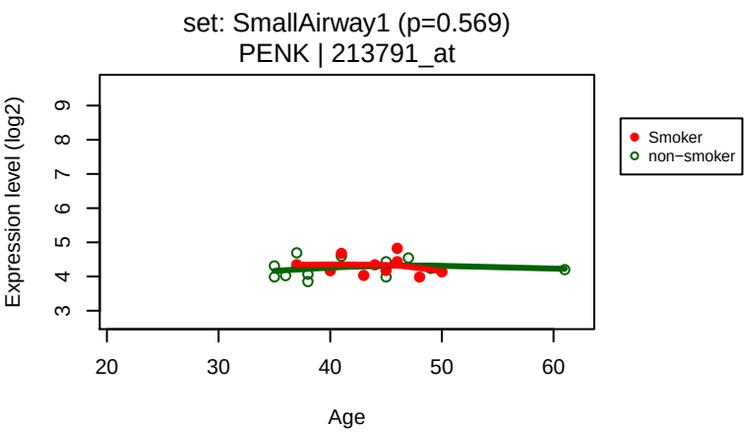
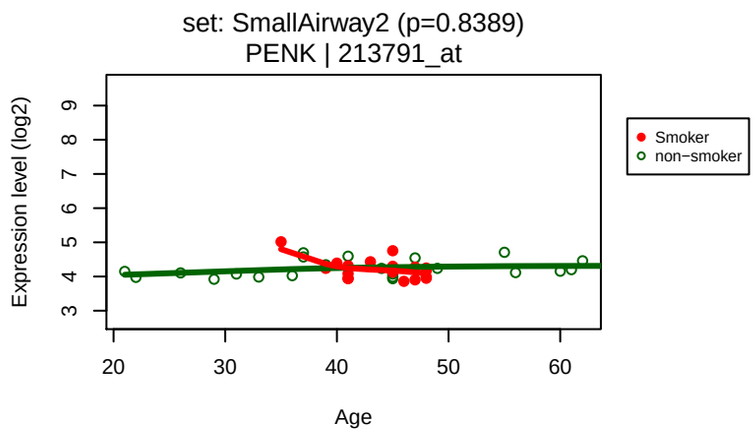
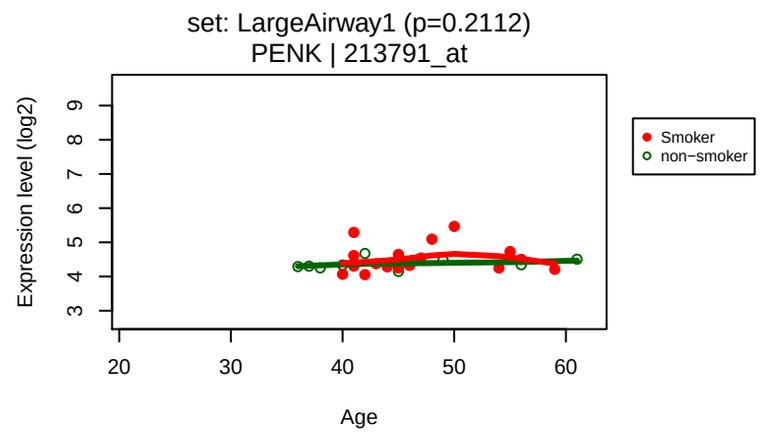

Gene expression of smokers and nonsmokers
gene: PENK | 213791_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.569
p=0.8389
p=0.2131
p=0.2112
p=0.9804
p=0.4462
p=0.235
p=0.9905
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

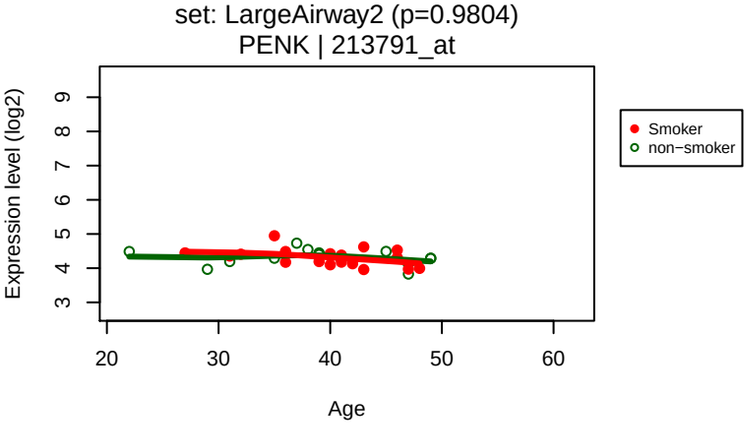

Gene expression ratio (smokers/nonsmokers)
gene: PENK | 213791_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.569
p=0.8389
p=0.2131
p=0.2112
p=0.9804
p=0.4462
p=0.235
p=0.9905
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

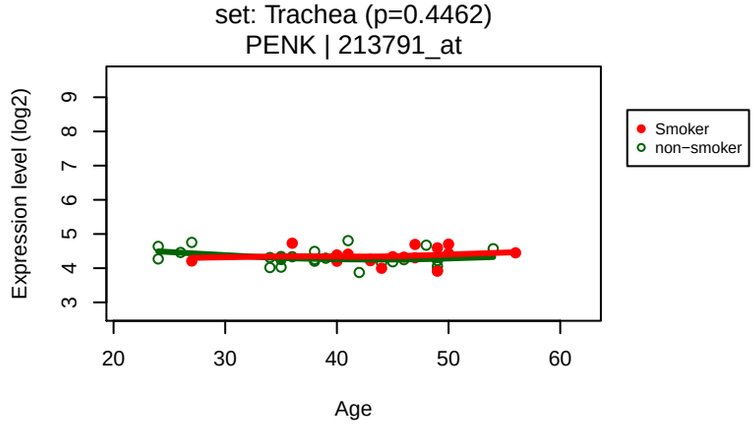

Gene expression ratio in Males
gene: PENK | 213791_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.941
p=0.7586
p=0.1122
p=0.9854
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PENK | 213791_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6032
p=0.8133
p=0.8568
p=0.9367
SmallAirway2
LargeAirway2
Trachea
Macrophage2

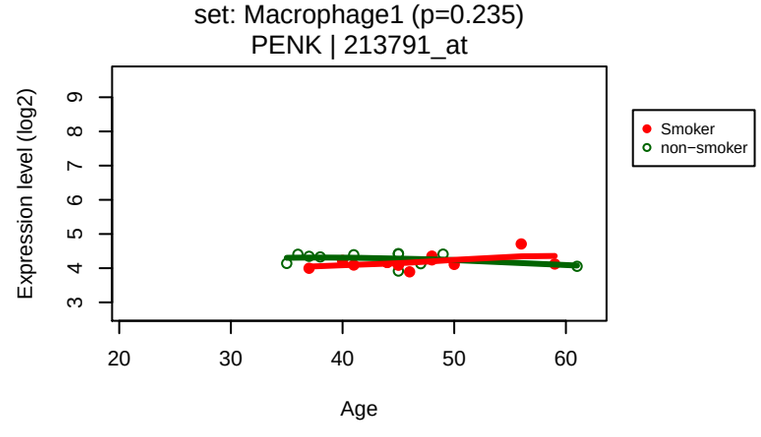

set: SmallAirway1 (p=0.569)
PENK | 213791_at
Expression level (log2)
Age
Smoker
non−smoker

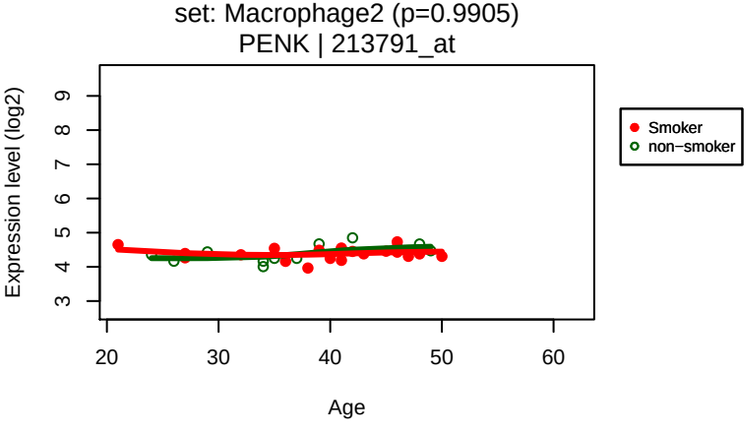

set: SmallAirway2 (p=0.8389)
PENK | 213791_at
Expression level (log2)
Age
Smoker
non−smoker

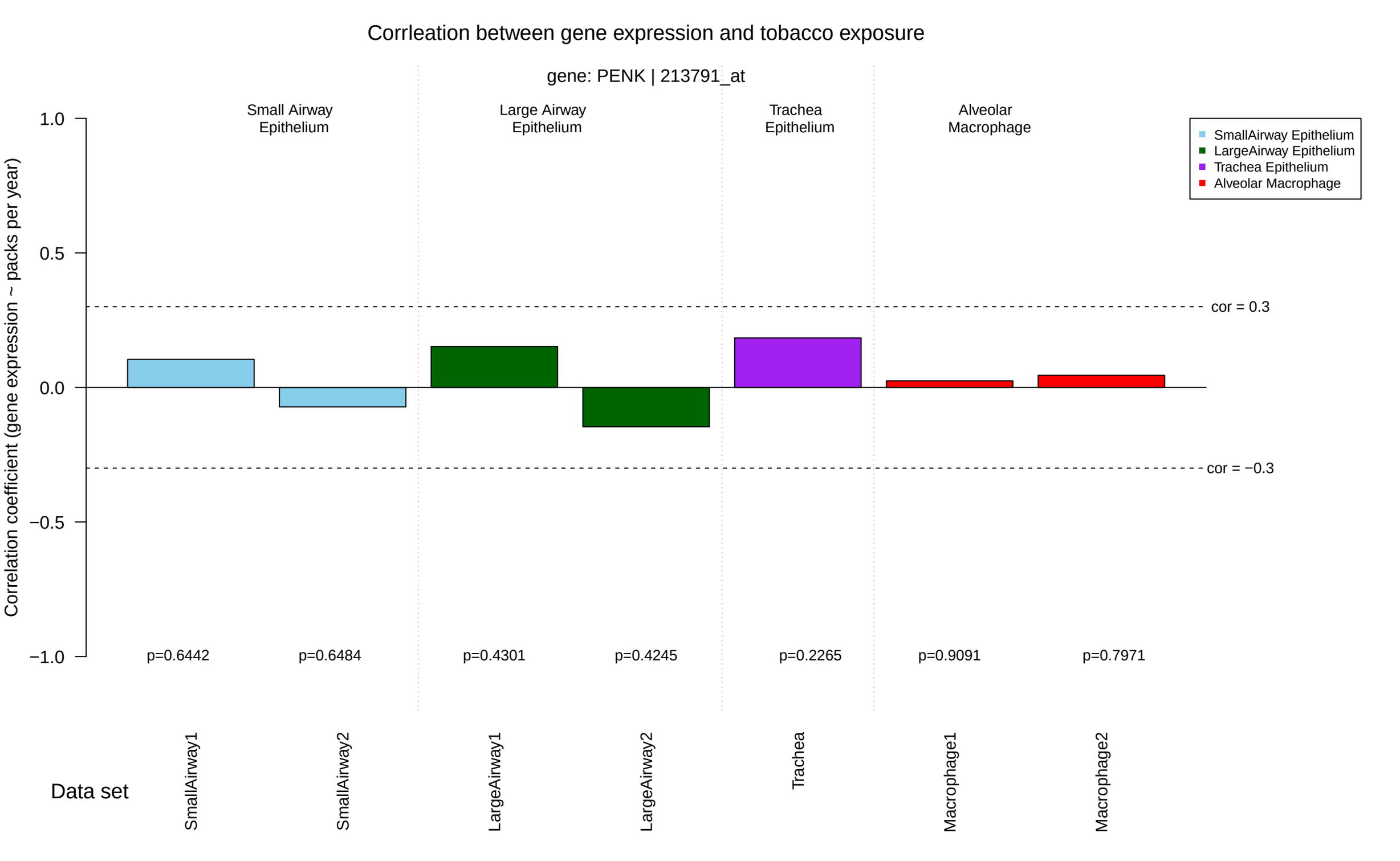

set: LargeAirway1 (p=0.2112)
PENK | 213791_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9804)
PENK | 213791_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4462)
PENK | 213791_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.235)
PENK | 213791_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9905)
PENK | 213791_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PENK | 213791_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6442
p=0.6484
p=0.4301
p=0.4245
p=0.2265
p=0.9091
p=0.7971
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2